Harnessing Bayesian Filters for Addressing Uncertain Variability in Chance-Constrained FDUC

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***Abstract*—A challenging problem in the low-inertia grid is to maintain frequency dynamic security while considering external uncertain variability (UV) caused by a large penetration of renewable energy sources. The proposed solution for this problem involves the utilization of a novel Bayesian-filter-derived System Frequency Response (BF-SFR) method and a chance-constrained frequency dynamic unit commitment (CCFDUC) model. Firstly, a Bayesian filtering approach is extended to capture the trajectory of the frequency derivation considering the UV assumption. In contrast to prior methodologies such as the analytical SFR method and time-discretized ones with free residual assumptions, BF-SFR not only calculate the change of frequency derivation like existing methods, but also offers an estimation of the posterior distribution of frequency derivation with different UV settings. Subsequently, the posterior distribution obtained in the BF-SFR framework through Sequential Importance Sampling (SIS) is reshaped as a collection of chance-constrained frequency dynamic constraints, and then integrated into the baseline UC problem. The linearization of chance-constrained nadir constraint is also designed and discussed in CCFDUC. The validity of the suggested models have been verified by the examination of multiple cases using a modified IEEE 6 bus test system and a 118 bus test system, in contrast to traditional UC model and state-of-art FDUC model.**

***Index Terms*—Bayesian filer, system frequency response, uncertain variability, unit commitment, chance constraints, converter-interfaced generators.**

NOMENCLATURE

Index and set

b	Index of bus, $b \in [1, 2, ..., NB]$
e	Index of Battery Electrical Storage System (BESS), $e \in [1, 2, ..., NE]$
g	Index of conventional generators, $g \in [1, 2..., NG]$
l	Index of transmission line, $l \in [1, 2, ..., NL]$
q	Index of hyperplane, $q \in [1, 2, ..., NQ]$
s	Index of stochastic scenario, $s \in [1, 2, ..., NS]$
t	Index of scheduling period, $t \in [1, 2, ..., NT]$
w	Index of wind farms, $w \in [1, 2, ..., NW]$
z	Index of unit's production cost blocks, $z \in [1, 2, ..., NZ]$

Parameters

Δf_{max}	Threshold of frequency nadir
Δp^E_{add}	Additional power of BESS
Δp^G_{add}	Additional power of conventional units
Δp^W_{add}	Additional power of wind farms
Δp_{step}	Mismatch power disturbance
Δt	discretized time step ($0.05s$ in this paper)
ϵ	Risk of confidence level
$\eta_e^{\text{dis/ch}}$	Discharging/charging efficiency of BESS e
$\gamma_g^{\text{ru/sd}}$	Ramp-up/ramp-down rate of conventional unit g
$\gamma_g^{\text{su/sd}}$	Start-up/shut-down capacity of conventional unit g
$\kappa_{l,b}$	Generator-shift-distribution-factor between line l and injected power located at bus b
$\omega_g^{(z)}$	Intercept of production cost block z of unit g
π_q	Probability of sampled residuals
π_s	Possibility of stochastic scenario s
$c_g^{\text{up/dw/nl}}$	Shut-up/shut-down/no-load cost of unit g
D_g	Damping of conventional generators
e^{VoLL}	Risk cost of Value-of-Lost-Load (VoLL)
F_g^z	Slope of cost block z of conventional unit g
F_g	Fraction representing total power generated by turbine of conventional reheated generators
K_g	Gain of conventional generators
$k_g^{(z)}$	Slope of operation cost block z of unit g
M	A big number
M_e	Inertia of BESS e
M_g	Inertia of conventional generators
M_w	Inertia of converters with virtual inertia control
$p_e^{\text{dis/ch, max}}$	Discharging/charging capacity of BESS e
$p_g^{\text{min/max}}$	Min/max output of conventional unit g
p_l^{max}	Capacity of transmission line l
$p_{w,s,t}$	Output of wind farm w at time t under scenario s
R_g	Droop of conventional generators
R_w	Droop of converters with droop control
$RoCoF_{\text{max}}$	Threshold of Rate-of-Change-of-Frequency(RoCoF)
t_0	Initial time of power disturbance
T_E	Aggregated time constant of BESS with droop control
T_e	Time constant of BESS e
$T_{G,t}$	Aggregated time constant of conventional units
T_g	Time constant of conventional generators
T_W	Aggregated time constant of converters
T_w	Time constant of converters
z_q	Auxiliary binary decision variables

Variables

$\Delta p_{d,s,t}$	Forced-load-curtailment at time t under scenario s
$\Delta p_{w,s,t}$	Wind curtailment at time t under scenario s
$D_{\text{equ},t}$	Aggregated equivalent damping at time t
$D_{G,t}$	Aggregated damping of conventional units at time t
$F_{G,t}$	Aggregated fraction of mechanical power generated by conventional reheated units at time t
$K_{G,t}$	Aggregated gain of conventional units at time t
$M_{\text{equ},t}$	Aggregated equivalent inertia at time t
$M_{E,t}$	Aggregated inertia of BESS at time t
$M_{G,t}$	Aggregated inertia of conventional units at time t
$M_{W,t}$	Aggregated inertia of converters at time t
$p_{e,s,t}^{\text{dis/ch}}$	Discharge/charge power of BESS e at time t under scenario s
$p_{g,s,t}$	Power of unit g at time t under scenario s
$p_{g,s,t}^{(z)}$	Output of conventional unit g at time t under scenario s corresponding to related cost block z
$R_{G,t}$	Aggregated droop of conventional units at time t
$R_{W,t}$	Aggregated droop of converters at time t
$u_{e,s,t}^{\text{ch}}$	Charge status of BESS at time t under scenario s
$u_{e,s,t}^{\text{dis}}$	Discharge status of BESS at time t under scenario s
$u_{g,t}$	Shut-up decision of conventional unit g at time t
$v_{g,t}$	Shut-down decision of conventional unit g at time t
$x_{g,t}$	Operation status of conventional unit g at time t

I. INTRODUCTION

THE intermittent of renewable energy give rise to UV concerns, which might have adverse effects on the frequency dynamic security in the low inertia grid. The consequences compassed in the analysis consist of an erroneous estimation of frequency trajectory, an excessive idealization of scheduling results and an inadequate response to the system in relation to imbalanced power disturbances. With the growing integration of converters for the purpose of providing frequency regulation services, there arises a need to develop a new and robust SFR model. Additionally, a scheduling strategy that addresses the challenges related to UV concerns must be redesigned.

Several research in the previous literature focused on modeling SFR in the presence of imbalanced power disturbances. The frequency dynamics in these experiments were calculated using a swing equation based on the concept of center-of-inertia (CoI). Ref. [1, 2] developed a CoI low-order equation for the system only considering conventional generators and then derived an analytical solution. Ref. [3–6] examined the effects of converter-interfaced generators employing different frequency control schemes. It is noteworthy to mention that the models described in [1–6] exhibit consistent mathematical formulations based on ordinary differential equations. Consequently, the solutions proposed by these works maintained comparable structures, with variations in the partial frequency dynamic parameters resulting from the incorporation of special frequency control schemes. The inadequacy of such methods arises from a lack of thorough examination of intricate control schemes for preserving the original SFR equation's structure. Motivated by this concern, another approach to represent SFR through time discretization is devised. Ref. [7] carried out a time-discretized approach that incorporates rate-of-change-of-frequency (RoCoF) replays. Ref. [8] expanded upon a time-discretized SFR form considering the frequency support provided by fly-wheel storage devices. Ref. [9, 10] developed a time-discretized SFR solution, taking into account the presence of converter-interfaced generators. This type of SFR method in [7–10] predict the trajectory of frequency dynamics under free residual assumptions, and the bottleneck of solving SFR considering UV assumptions remains unresolved. Moreover, accurate estimation of the posterior probabilistic distribution of frequency derivation is challenging.

Moreover, frequency dynamics in electrical power system operation & control are extended through the implementation of FDUC. the aim of FDUC is to integrate frequency dynamic requirements obtained from SFR into the baseline UC problem. Specially, FDUC can be classified into two types based on their adherence to integrated frequency dynamic restrictions. $i)$ The crucial characteristic requirement of frequency dynamics, i.e., RoCoF and nadir, is reshaped into a set of convex constraints in FDUCs, as specified in time intervals [11–19]. $ii)$ The alteration in frequency over a predetermined simulated time span (e.g., 0s - 60s) is redefined as a series of obligatory constraints that are integrated into the UC problem, as outlined in [8, 20, 21]. In contrast to the previous category, the latter category is characterized by a higher level of accuracy yet with more complexity. While there has been significant research on the participation of conventional generators in frequency regulation services for supporting frequency, there is a dearth of studies that focus on the probabilistic of frequency support offered by converter-interfaced generators.

This research introduces a novel probabilistic Bayesian filter architecture that encodes the SFR considering UV hypothesis, driven by the aforementioned concerns. BF-SFR captures the interaction of UV on the frequency dynamic response. It not only allows for an estimation of frequency dynamic trajectory but also provides the posterior PDF of frequency derivation. So, BF-SFR offers a high flexibility in examining frequency dynamics and UV factors. Subsequently, this study analysis the influence of frequency dynamic requirement derived from BF-SFR inside the FDUC framework. Additionally, an enhanced CCFDUC model is effectively revised.

Two contributions can be succinctly summarized as follows: $i)$ a novel method called the BF-SFR is suggested for modeling SFR. It examines the SFR that considers the effect of the UV issue, and accurate estimation of posterior probabilistic distribution concerning the frequency derivation is analyzed. $ii)$ a CCFDUC model is extended, which incorporates frequency dynamics using the BF-SFR method. A chance-constraint form of frequency dynamic requirement related to nadir is generated and reformulated. Besides, this study investigates the effects of modified frequency limitations, implemented with varying UV configurations, on the resulting scheduled commitments.

The reminder of this paper is organized: Section II introduces a Bayesian filter framework for encoding SFR and derives the solution of BF-SFR. Section III constructs a convex set of chance-constraint frequency nadir constraints as well as RoCoF constraint and FCR constraint. Subsequently, a CCFDUC model is presented in Section IV. Section V verifies the effectiveness of BF-SFR and CCFDUC in a modified IEEE 14 bus test system and IEEE 118 bus test system, and Section VI provides a summary of primary findings and conclusions presented in this research.

II. Modeling BF-SFR through Bayesian-Filter

Considering a low-inertia grid comprised of conventional generators, wind farms and BESS, it is seen that the effect of homogeneous generators with comparable control schemes can be seen as an aggregated CoI block. Define aggregated parameters $\{H_G, D_G, K_G, F_G, R_G, T_G\}$, $\{K_W, R_W, M_W, D_W, T_W\}$ and $\{T_E, R_E\}$ represent adjusted output produced by CoI conventional generator block, CoI wind farm block and CoI BESS block. Further details on the derivation from frequency parameters of individual units to aggregated unit block can be found in [11**?** –16]. The transfer function of SFR concerning a occurred step-type power disturbance $\Delta p_{\text{step}}(s) = \Delta p_{step}/s$ can be rewritten as,

$$\begin{aligned}(M_G s + D_G)\,\Delta f(s) = (\,1/s\ \Delta p_{\text{step}}\,) - \\ \underbrace{(\,K_G/R_G\,)\,[\,(1+F_G T_G s)/(1+T_G s)\,]\,\Delta f(s)}_{\text{CoI conventional unit}} - \\ \underbrace{(\,1/R_W\,)\,(\,1/1+T_W s\,)\,\Delta f(s)}_{\text{CoI droop-controlled converter-interfaced unit}} - \underbrace{(\,1/R_E\,)\,(\,1/1+T_E s\,)\,\Delta f(s)}_{\text{CoI BESS}}\end{aligned} \tag{1}$$

The $t-$ domain mathematical form of (1) can be transferred by utilizing the inverse Laplace transformation,

$$M_{\text{equ}}\Delta f'(t) + D_{\text{equ}}\Delta f(t) = \Delta p(t) \tag{2a}$$

$$\Delta p(t) = \Delta p_{\text{step}} - \Delta p^G_{\text{add}}(t) - \Delta p^W_{\text{add}}(t) - \Delta p^E_{\text{add}}(t) \tag{2b}$$

$$M_{\text{equ}} = M_G + M_W \tag{2c}$$

$$D_{\text{equ}} = D_G + D_W + R_W^{-1} + R_E^{-1} \tag{2d}$$

$$\begin{aligned}\Delta p^G_{\text{add}}(t) = K_G F_G R_G^{-1}\Delta f(t) - \\ K_G(1-F_G)R_G^{-1}e^{-\frac{1}{T_G}t}\int_0^t e^{\frac{1}{T_G}\kappa}\Delta f'(\kappa)d\kappa\end{aligned} \tag{2e}$$

$$\Delta p^W_{\text{add}}(t) = \left(R_W^{-1}\right)\Delta f(t) \tag{2f}$$

$$\Delta p^E_{\text{add}}(t) = \left(R_E^{-1}\right)\Delta f(t) \tag{2g}$$

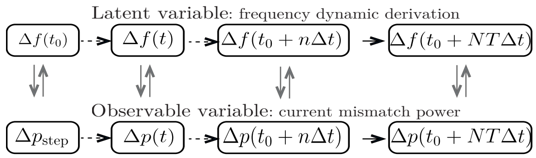

Figure 1. The time-discretized status recurrence process of frequency derivation in the proposed BF-SFR method

By replacing $\Delta f'(t)|_{t=t_0+n\Delta t}$ as its equivalent discretized form $[\Delta f(t_0+n\Delta t)-\Delta f(t_0+(n-1)\Delta t)]/\Delta t$, the time-discretized form of $\Delta f(t)$ can be estimated as below,

$$\Delta f(t_0+(n+1)\Delta t) = \Delta f(t_0+n\Delta t) + M_{\text{equ}}^{-1}[\Delta p(t_0+n\Delta t) - D_{\text{equ}}\Delta f(t_0+n\Delta t)] \tag{3a}$$

$$\Delta p(t_0+n\Delta t) = \Delta p_{\text{step}} - \Delta p_{\text{add}}^{G}(t_0+n\Delta t) - \Delta p_{\text{add}}^{W}(t_0+n\Delta t) - \Delta p_{\text{add}}^{E}(t_0+n\Delta t) \tag{3b}$$

$$\Delta p_{\text{add}}^{G}(t_0+n\Delta t) = K_G R_G^{-1}\Delta f(t_0+n\Delta t) - \left(K_G(1-F_G)R_G^{-1}e^{-\frac{1}{T_G}(t_0+n\Delta t)}\right) \times \left(\sum_{i=0}^{n} e^{\frac{1}{T_G}(t_0+i\Delta t)}[\Delta f(t_0+i\Delta t) - \Delta f(t_0+(i-1)\Delta t)]\right) \tag{3c}$$

$$\Delta p_{\text{add}}^{W}(t_0+n\Delta t) = \left(R_W^{-1}\right)\Delta f(t_0+n\Delta t) \tag{3d}$$

$$\Delta p_{\text{add}}^{E}(t_0+n\Delta t) = \left(R_E^{-1}\right)\Delta f(t_0+n\Delta t) \tag{3e}$$

Eq. (3a) represents the evolution of $\Delta f(t)$ between adjacent discretized steps; Eq. (3b) describes the evolution of mismatch power. Eqs. (3c) to (3e) represent additional regulated outputs produced by aggregated conventional generators, aggregated wind farms and aggregated BESS.

Previous studies generally adhere to a free residual assumption and has provided evidence that $\Delta f(t)$ can be achieved by either an analytical closed-form solution [1–6] or a time-discretized solution [7–10]. In contrast to previous studies that have been discussed, the BF-SFR approach incorporates the influence of residual representing UV issue. The significance of UV is understood in two main aspects: i) the intermittent characteristic of converter-interfaced generators participating in frequency support, leading to the inevitable residual relative to the ideal value; ii) potential measured residuals due to the aging behavior or aberrant operation of electrical devices. By formulating the SFR considering residual factors as a hidden Markov decision process, it becomes possible to estimate the posterior distribution of $\Delta f(t)$ using a recursive mathematical formulation.

Let $\mathcal{P}(\Delta f(t_0+n\Delta t), \mu_1, \sigma_1)$ represents the posterior PDF of $\Delta f(t_0+n\Delta t)$ at the $n\text{–}th$ discretized step; $\mathcal{P}(\Delta p(t_0+n\Delta t), \mu_2, \sigma_2)$ denotes the posterior PDF of $\Delta p(t_0+n\Delta t)$ at the $n\text{–}th$ discretized step. $\langle \mu_1, \sigma_1\rangle$, $\langle \mu_2, \sigma_2\rangle$, respectively, denote the expectation and variance of residuals in the observation process and measured process. $\mathcal{P}(\Delta f(t_0+(n+1)\Delta t) \mid \Delta p(t_0:t_0+(n+1)\Delta t), u_1, \sigma_1)$ to denote an conditional distribution of $\mathcal{P}(\Delta f(t_0+(n+1)\Delta t), u_1, \sigma_1)$, It describes that the posterior of $\mathcal{P}(\Delta f(t_0+(n+1)\Delta t), u_1, \sigma_1)$ can be fully inferred after obtaining all measured sequences $\Delta p(t_0:t_0+(n+1)\Delta t) = \{\Delta p_t \mid t \in [t_0, t_0+(n+1)\Delta t]\}$. Its mathematical formulation can be derived as Eq. (4) according to the Bayesian theorem.

In Eq. (4), $\mathcal{P}(\Delta f(t_0+(n+1)\Delta t \mid \Delta p(t_0:t_0+(n+1)\Delta t), u_1, \sigma_1)$ can be indirectly calculated by estimating its prior form $\mathcal{P}(\Delta f(t_0+(n+1)\Delta t) \mid \Delta p(t_0:t_0+n\Delta t), u_1, \sigma_1)$ and a related normalized constant $\mathcal{P}(\Delta p(t_0+(n+1)\Delta t) \mid \Delta f(t_0+(n+1)\Delta t), u_2, \sigma_2)$. Their mathematical forms can be found in Eq. (5).

$$\mathcal{P}(\Delta f(t_0+(n+1)\Delta t) \mid \Delta p(t_0:t_0+n\Delta t), u_1, \sigma_1) = \int \{\mathcal{P}(\Delta f(t_0+(n+1)\Delta t) \mid \Delta f(t_0+n\Delta t), \mu_1, \sigma_1)\mathcal{P}(\Delta f(t_0+n\Delta t) \mid \Delta p(t_0:t_0+n\Delta t), u_1, \sigma_1)\} \cdot d\Delta f(t_0+n\Delta t) \tag{5a}$$

$$\mathcal{P}(\Delta p(t_0+(n+1)\Delta t) \mid \Delta p(t_0:t_0+n\Delta t), u_2, \sigma_2) = \int \{\mathcal{P}(\Delta p(t_0+(n+1)\Delta t) \mid \Delta f(t_0+(n+1)\Delta t)\ u_2, \sigma_2)\ \mathcal{P}(\Delta f(t_0+(n+1)\Delta t) \mid \Delta p(t_0:t_0+n\Delta t), u_2, \sigma_2)\} d\Delta f(t_0+(n+1)\Delta t) \tag{5b}$$

Next, let the initial time $t_0 = 0$ and replace $\langle \Delta f(t_0+n\Delta t)\rangle$, $\langle \Delta p(t_0+n\Delta t)\rangle$ with their abbreviated forms $\langle \Delta f(n), \Delta p(n)\rangle$, Eq. (5) can be simplified as follows,

$$\mathcal{P}(\Delta f(n+1) \mid \Delta p(0:n), u_1, \sigma_1) = \int \{\mathcal{P}(\Delta f(n+1) \mid \Delta f(n), u_1, \sigma_1)\,\mathcal{P}(\Delta f(n) \mid \Delta p(0:n), u_1, \sigma_1)\} d\Delta f(n+1) \tag{6a}$$

$$\mathcal{P}(\Delta p(n+1) \mid \Delta p(0:n), u_2, \sigma_2) = \int \{\mathcal{P}(\Delta p(n+1) \mid \Delta f(n+1)\ u_2, \sigma_2)\,\mathcal{P}(\Delta f(n+1) \mid \Delta p(0:n), u_2, \sigma_2)\} d\Delta f(n+1) \tag{6b}$$

To calculate the posterior function of $\mathcal{P}(\Delta f(n+1) \mid \Delta p(0:n), u_1, \sigma_1)$ and $\mathcal{P}(\Delta p(n+1) \mid \Delta p(0:n), u_2, \sigma_2)$, the operator $\int(\cdot)$in (5) can be approximately estimated by an expectation of a certain number of sampled data (or called $filters$), i.e., $\int \varpi[\Delta f(n)]d\Delta f(n) \approx \frac{1}{N}\sum_{n=1}^{N} \varpi(\lambda_n)\lambda_n$ where ϖ enforces associated integrand function in (6) and λ_n represents possible $filters$ generated from the prior of $\Delta f(n)$. Hence, (6) can be rewritten as,

$$\mathcal{P}(\Delta f(n+1) \mid \Delta p(0:n), u_1, \sigma_1) = N^{-1}\sum_{n=1}^{N} \mathcal{P}(\Delta f(n+1) \mid \Delta f(n), u_1, \sigma_1)\,\lambda_n' \tag{7a}$$

$$\lambda_n' \sim \mathcal{P}(\Delta f(n) \mid \Delta p(0:n), u_1, \sigma_1) \tag{7b}$$

$$\mathcal{P}(\Delta p(n+1) \mid \Delta p(0:n), u_2, \sigma_2) = N^{-1}\sum_{n=1}^{N} \mathcal{P}(\Delta p(n+1) \mid \Delta f(n+1), u_2, \sigma_2)\,\lambda_n'' \tag{7c}$$

$$\lambda_n'' \sim \mathcal{P}(\Delta f(n+1) \mid \Delta p(0:n), u_2, \sigma_2) \tag{7d}$$

where λ', λ_n'', respectively, represent random filters following associated integrand functions in (7b) and (7d). In combination with (7a) and (7c), the posterior probabilistic assessment of $\Delta f(n+1)$ concerning $\mathcal{P}(\Delta f(n+1))$ can be carried out with the below steps:

i) *Bayesian inference* in the prior stage, here, (7a) is used to generate *filters* to capture the likely value of $\Delta f(n+1)$;

ii) *Bayesian predication* in the prediction stage, (4) and (7c) is used to update the posterior of $\Delta f(n+1)$. Since λ', λ_n'' cannot directly generated from $\mathcal{P}(\Delta f(n) \mid \Delta p(0:n), u_1, \sigma_1)$ and $\mathcal{P}(\Delta f(n+1) \mid \Delta p(0:n), u_2, \sigma_2)$ because their functions are unknown, Therefore, sequential Important Sampling (SIS) method is used to calculate the integrand functions. Specially, $\mathcal{P}(\Delta f(n) \mid \Delta p(0:n), u_1, \sigma_1)$ at the $n-th$ prior stage is took as an example to describe the

$$\mathcal{P}(\Delta f(t_0+(n+1)\Delta t) \mid \Delta p(t_0:t_0+(n+1)\Delta t), u_1, \sigma_1) = \frac{\mathcal{P}(\Delta f(t_0+(n+1)\Delta t) \mid \Delta p(t_0:t_0+n\Delta t), u_1, \sigma_1)\,\mathcal{P}(\Delta p(t_0+(n+1)\Delta t) \mid \Delta f(t_0+(n+1)\Delta t), u_1, \sigma_1)}{\mathcal{P}(\Delta p(t_0+(n+1)\Delta t) \mid \Delta p(t_0:t_0+n\Delta t), u_2, \sigma_2)} \tag{4}$$

Table I
THE SIMULATION OF BF-SFR THROUGH SIS ALGORITHM

NO.	Calculation processes
1	**Initialize**: Discretized step $\Delta t = 0.05s$; Initialized time $t_0 = 0s$ Counter n = 0; Filter number NT = 1000
2	**For** $n = 0$, **Then**
3	**Let** $t = t_0 + \Delta t \times n$, **Then**
4	**Inference process**: Generating possible (instant) filters $(\Delta f)^{(n)}\,(t_0 + (n+1)\Delta t)$ from proposed distribution: $\lambda'(n+1) \sim \mathcal{P}(\Delta f(t_0 + (n+1)\Delta t) \vert \Delta p(t_0 : t_0 + n\Delta t), \mu_1, \sigma_1)$ in (6a);
5	**Prediction process**: Generating possible observation filters $(\Delta p)^{(n)}\,(t_0 + (n+1)\Delta t)$ from normalized distribution: $\lambda''(n+1) \sim \mathcal{P}(\Delta p(t_0 + (n+1)\Delta t) \vert \Delta p(t_0 : t_0 + n\Delta t), \mu_2, \sigma_2)$ in (6b);
6	**Updating weight coefficients:** Calculating γ' and γ'' through (9), and Re-estimate original filters $(\Delta f)^{(n)}\,(t_0 + (n+1)\Delta t)$ via (4)
7	**End Let**
8	**End For**

calculating process using SIS, the case of the latter (posterior) stage concerning $\mathcal{P}\left(\Delta f(n+1)\,\vert\,\Delta p\,(0:n)\,, u_2, \sigma_2\right)$ is in a similar solution.

Let $\iota(n) = \mathcal{P}\left(\Delta f(n)\,\vert\,\Delta p\,(0:n)\,, u_2, \sigma_2\right), \iota(n) = \left({}^{\iota(n)}/_{\lambda(n)}\right) \cdot \lambda(n)$ where $\iota(n)$ is the original probabilistic density function of posterior distribution at $t{=}n\Delta t$; $\lambda(n)$ enforces a proposed function. $\gamma(n) = {}^{\iota(n)}/_{\lambda(n)}$ represents a likelihood coefficient. Hence, the $filters$ following $\iota(n)$ can be alternatively sampled from $\lambda(n)$ with a related weight coefficient $\gamma(n)$, i.e., $\iota(n) = \lambda(n)\gamma(n)$.

$$\gamma(n{+}1) = \frac{\iota(\Delta f(n{+}1) \vert \Delta p(0{:}n{+}1), \mu_1, \sigma_1)}{\lambda(\Delta f(n{+}1) \vert \Delta p(0{:}n{+}1), \mu_2, \sigma_2)} = \gamma(n) \times \left(\frac{\iota(\Delta f(n{+}1) \vert \Delta f(t), \mu_1, \sigma_1)\iota(\Delta p(n{+}1) \vert \Delta f(n{+}1), \mu_2, \sigma_2)}{\lambda(\Delta f(n{+}1) \vert \Delta f(t), \mu_1, \sigma_1)}\right) \tag{8}$$

Substituting $\iota(n+1) = \lambda(n+1)\gamma(n+1), \iota(n) = \lambda(n)\gamma(n)$ into (4), $\iota(n+1)$ can be recursively calculated through (8). In (8), another critical issue to be reviewed is the selection of the proposed distribution. Here, the recursive transfer distribution $\iota(\Delta f(n+1) \vert \Delta f(n), \mu_1, \sigma_1)$ can be availability used to simplify the calculation process. Accordingly, the updating weight (8) can be reduced as,

$$\gamma(n+1) = \gamma(n)\iota(\Delta p(n+1) \vert \Delta f(n+1), \mu_2, \sigma_2) \tag{9}$$

In (9), the likelihood rate in each iteration can be recursively derived from $\iota(\Delta p(n+1) \vert \Delta f(n+1), \mu_2, \sigma_2)$. On this basis, let $\langle \Delta f^{(n)}(t + (n+1)\Delta t), (\Delta p)^{(n)}\,(t + (n+1)\Delta t) \rangle$ define the sampled *filters* from the pro-selected proposed function $\lambda'(n{+}1)$ in the inference stage and $\lambda''(n{+}1)$ in the predication stage; $\langle \gamma'(n+1), \gamma''(n+1) \rangle$ represent the likelihood rate at $t{=}t_0 + (n+1)\Delta t$ at these two stages. Table I describes the Pseudo-Code of the proposed BF-SFR method.

III. FREQUENCY CONSTRAINT REFORMULATION

This section examines the represented criteria of frequency dynamic requirements, including RoCoF, nadir, and frequency containment reserve (FCR). These needs are reevaluated and integrated into the restrictions considered in UC problems. Fig. 2 provides a visual representation of the physical significance concerning these feature criteria. Remarkably, the reconstructed chance-constrained frequency nadir constraint is derived from the novel BF-SFR approach presented in section II, other constraints related to RoCoF and FCR may be found in references [8, 11–16, 20**?** , 21].

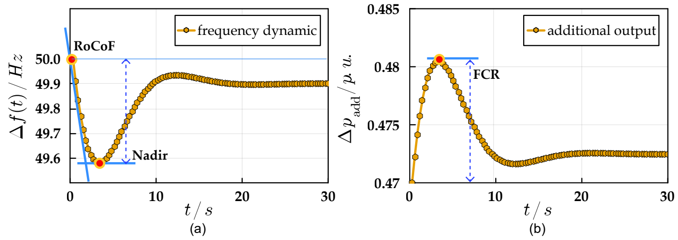

Figure 2. Three frequency dynamic requirements: $RoCoF$, $nadir$ and FCR

A. $RoCoF$ constraint

To mitigate rapid changes in the inertia response, the change of RoCoF, indicated as $\Delta f'(t)$, is necessary. It is required that $\Delta f'(t)$ does not exceed the maximum rate of change of frequency, denoted as $RoCoF_{\text{max}}$, *i.e.,* $\Delta f'(t) \leq RoCoF_{\text{max}}$. Since $\max\{\Delta f'(t)\} = {}^{\Delta p_{\text{step}}}/_{M_{\text{equ}}} : M_{\text{equ}} = M_G + M_W$, [22], associated $RoCoF$ constraint can be obtained,

$$\begin{aligned} &\Delta p_{\text{step}}/M_{\text{equ}} \leq RoCoF_{\text{max}} \\ &\iff M_G \geq \Delta p_{\text{step}}/RoCoF_{\text{max}} - M_W, \forall g, t \\ &\iff \sum\nolimits_{g=1}^{NG} M_g x_{g,t} \geq \frac{\Delta p_{\text{step}}}{RoCoF_{\text{max}}} - \sum\nolimits_{w=1}^{NW} M_w, \forall g, t \end{aligned} \tag{10}$$

B. FCR constraint

To guarantee sufficient frequency reserve provided by grid-connected units, it is imperative that both conventional generators, converters and BESS are capable of serving as reserves during imbalanced disturbances, [15, 23]. In combination with (3), the binding of the additional power $\langle \Delta p^G_{\text{add}}, \Delta p^W_{\text{add}}, \Delta p^E_{\text{add}} \rangle$ from different units is obtained, *i.e.*., $\sum_{g=1}^{NG} p_{g,s,t} \geq \Delta p^G_{\text{add}}(t)$; $\sum_{w=1}^{NG} p_{w,s,t} \geq \Delta p^W_{\text{add}}(t)$ and $\sum_{e=1}^{NE} p_{e,s,t} \geq \Delta p^E_{\text{add}}(t), \forall t, s$. By replacing $\langle \Delta p^G_{\text{add}}(t), \Delta p^W_{\text{add}}(t), \Delta p^E_{\text{add}}(t) \rangle$ with their analytical formulations in Eqs. (3c) to (3e), another equivalent constraint regarding FCR can be derived,

$$\sum\nolimits_{w=0}^{NW} (p^{\text{max}}_w - p_{w,s,t}) \geq R^{-1}_W \Delta f(t) \tag{11a}$$

$$\sum\nolimits_{e=0}^{NE} \left(p^{\text{max}}_e - p^{\text{dis}}_{e,s,t}\right) \geq R^{-1}_E \Delta f(t) \tag{11b}$$

$$\sum\nolimits_{g=0}^{NG} (p^{\text{max}}_g - p_{g,s,t}) \geq K_G R^{-1}_G \Delta f(t_0 + n\Delta t) - K_G(1 - F_G) R^{-1}_G e^{-\frac{1}{T_G}(t_0 + n\Delta t)} \left(\sum_{i=0}^{n} e^{\frac{1}{T_G}(t_0 + i\Delta t)} \Delta f'(t_0 + i\Delta t)\right) \Delta t \tag{11c}$$

Note that the upper bound of $\Delta f(t)$ in (11) is predetermined as Δf_{max}, therefore, a concise form of (11a) and (11b) can be calculated by replacing $\Delta f(t)$ with Δf_{max},

$$\sum\nolimits_{w=0}^{NG} (p^{\text{max}}_w - p_{w,s,t}) \geq \Delta f_{\text{max}}/R_W\ , \forall s, t \tag{12a}$$

$$\sum\nolimits_{e=0}^{NE} \left(p^{\text{max}}_e - p^{\text{dis}}_{e,s,t}\right) \geq \Delta f_{\text{max}}/R_E\ , \forall s, t \tag{12b}$$

In the context of the FCR constraint (11c) pertaining to conventional generators, the right-hand side of (11c) can be rephrased as,

$$\begin{aligned} &K_G R^{-1}_G \Delta f(t_0 + n\Delta t) - \left(K_G(1 - F_G) R^{-1}_G RoCoF_{\text{max}}\right)(n\Delta t) \\ &\leq K_G R^{-1}_G \Delta f(t_0 + n\Delta t) - \left(K_G(1 - F_G) R^{-1}_G RoCoF_{\text{max}}\right) \\ &\quad \left(e^{-\frac{1}{T_G}(t_0 + n\Delta t)} \sum\nolimits_{i=0}^{n} e^{\frac{1}{T_G}(t_0 + i\Delta t)}\right) \Delta t \\ &\leq K_G R^{-1}_G \Delta f(t_0 + n\Delta t) - K_G(1 - F_G) R^{-1}_G \times \\ &\quad e^{-\frac{1}{T_G}(t_0 + n\Delta t)} \left(\sum\nolimits_{i=0}^{n} e^{\frac{1}{T_G}(t_0 + i\Delta t)} \Delta f'(t_0 + i\Delta t)\right) \Delta t \\ &\leq K_G R^{-1}_G \Delta f(t_0 + n\Delta t) \leq K_G R^{-1}_G \Delta f_{\text{max}} \end{aligned} \tag{13}$$

By substituting the maximum limit of changeable output in (13) into (14), a relaxed constraint on the FCR for conventional generators is calculated,

$$\sum\nolimits_{g=1}^{NG} \left(p^{\text{max}}_g - p_{g,s,t}\right) \geq K_G R^{-1}_G \Delta f_{\text{max}}, \forall s, t \tag{14}$$

C. $(chance-constraint)$ $nadir$ $constraint$ $reformulation$

To avoid the occurrence of $\Delta f(t)$ beyond its predetermined threshold $\Delta f_{\max}$, it is essential to impose nadir constraint on the potential fluctuation of $\Delta f_{\text{nadir}} = \max\{\Delta f(t)\}$, ensuring that Δf_{nadir} does not exceed $\Delta f_{\max}$, *i.e.,* $\Delta f_{\text{nadir}} \leq \Delta f_{\max}$. Note that the probabilistic density distribution of $\mathcal{P}(\Delta f(t))$ may be effectively determined using the BF-SFR method, as a result, a chance-constrained nadir constraint can be formulated as follows:

$$\mathcal{P}\left(\Delta f_{\text{nadir}} \leq \Delta f_{\max}\right) \geq 1-\epsilon \tag{15}$$

where ϵ is the confidence risk level associated with the chance constraint (15); Δf_{nadir} is closely dependent with the initial imbalance disturbance Δp_{step}, aggregated frequency parameters of various grid-connected units $\langle M_G, D_G, R_G, K_G, T_G, R_W, R_E\rangle$ and external UV characteristic $\langle \mu_1, \sigma_1, \mu_2, \sigma_2\rangle$. For simplification purpose, let $\wp = \langle M_G, D_G, R_G, K_G, T_G, R_W, R_E\rangle$ and $\zeta = \langle \mu_1, \sigma_1, \mu_2, \sigma_2\rangle$, and then, Δf_{nadir} can be rewritten a nonlinear mapping function $\Psi(\wp,\zeta)$, *i.e.,* $\Delta f_{\text{nadir}} = \Psi(\wp,\zeta)$. Furthermore, a Sampled Average Approximate (SAA) method is used to construct the deterministic convex formulation of (15) which can be readily identified by any off-the-shelf MIP solvers and subsequently integrate it into UC. To achieve that, by introducing a set of binary indicator variable $z_q \in \{0,1\} : q \in \{1,2,\ldots,NQ\}$ and using the Big-M method [14, 15], Eq. (15) can be equivalently reformulated with a set of linearized constraints as follows,

$$\Psi\left(\wp,\hat{\zeta}^{(q)}\right) \leq \Delta f_{\max} + M z_q, \sum\nolimits_{q=1}^{NQ} \pi_q z_q \leq \epsilon, \forall q \tag{16a}$$

where (16) represents an approximately mix-integer linearized constraints of (15). $z_q = 0$ indicates an active constraint where the $nadir$ must be required for the predefined uncertain variability condition; $z_q = 1$ describes an inactive constraint where the (15) at the predefined sampled uncertain condition q could be ignored. Accordingly, the redesigned chance constraint (16) has been reshaped as an equivalence of restricting the number of $z_q, 1 \leq q \leq NQ$ to be ones. It is important to mention that the mathematical reformulation of the similar type constraint (15) are not limited to the form (16). By directly introducing z_q but no longer introducing M, a bi-linear type mixed integer reformulation of (15) is also feasible, that is,

$$\boxed{\left(\Psi\left(\wp,\hat{\zeta}^{(q)}\right) - \Delta f_{\max}\right)(1-z_q) \leq 0}, \sum\nolimits_{q=1}^{NQ} \pi_q z_q \leq \epsilon, \forall q \tag{17a}$$

The boxed term in Eq. (17) can be rewritten as $\Psi(\wp,\hat{\zeta}^{(q)}) - z_q\Psi(\wp,\hat{\zeta}^{(q)}) - (1-z_q)\Delta f_{\max} \leq 0, \forall q$ where the nonlinear term $\Psi(\wp,\hat{\zeta}^{(q)})$ needs to be further linearized and the bi-linear term $\langle z_q \times \Psi(\wp,\hat{\zeta}^{(q)})\rangle$ should be reformulated through McCormick linearization method. For pretending the resulting constraint from being excessive complexity, (16) is utilized to approximate (15). Note that (16) cannot be directly incorporated into UC due to the nonlinear property of $\Psi(\wp,\hat{\zeta}^{(q)})$, therefore, a "max-affine" PWL method, [6, 24, 25], is implemented to estimate the original value of $\Psi(\wp,\hat{\zeta}^{(q)})$.

Let $\hat{\Psi}(\wp,\hat{\zeta}^{(q)}) = \max\left(\alpha_{\iota,1}^{(q)}\tilde{H}_{\text{equ}}^{(q)} + \alpha_{\iota,2}^{(q)}\tilde{D}_{\text{equ}}^{(q)} + \alpha_{\iota,3}^{(q)}\left(\frac{\tilde{F}_G^{(q)}}{\tilde{R}_G^{(q)}}\right) + \alpha_{\iota,4}^{(q)}\right)$ describe the approximated fitting function of $\hat{\Psi}(\wp,\hat{\zeta}^{(q)})$ given a stochastic uncertain realization $\hat{\zeta}^{(q)}$; $\langle \tilde{H}_{\text{equ}}^{(q)}, \tilde{D}_{\text{equ}}^{(q)}, \tilde{F}_G^{(q)}, \tilde{R}_G^{(q)}$ denotes a possible set of aggregated frequency parameters with the respect of $\hat{\zeta}^{(q)}$. In such a situation, $\hat{\Psi}(\wp,\hat{\zeta}^{(q)})$ can be evaluated by observing the maximum value of constructed hyperplane $\Theta_{\iota\in\{1,2,3,4\}}^{(q)}$ where $\Theta_\iota^{(q)} = \alpha_{\iota,1}^{(q)}\tilde{H}_{\text{equ}}^{(q)} + \alpha_{\iota,2}^{(q)}\tilde{D}_{\text{equ}}^{(q)} + \alpha_{\iota,3}^{(q)}\left(\frac{\tilde{F}_G^{(q)}}{\tilde{R}_G^{(q)}}\right)$. Particularly, the selection of fitting parameters $\langle \alpha_{\iota,1}^{(q)}, \alpha_{\iota,2}^{(q)}, \alpha_{\iota,3}^{(q)}, \alpha_{\iota,4}^{(q)}\rangle$ can be solved by minimizing the lose function defined by $\vartheta_{\text{loss}} = \hat{\Psi}(\wp,\hat{\zeta}^{(q)}) - \Psi(\wp,\hat{\zeta}^{(q)})$. As such, the below lists a detailed optimization model for the calculation of fitting parameters.

$$\begin{aligned}\min\sum\nolimits_{r=1}^{NR}&\left(\max\left(\alpha_{\iota,1}^{(q)}\tilde{H}_{\text{equ}}^{(q)} + \alpha_{\iota,2}^{(q)}\tilde{D}_{\text{equ}}^{(q)} + \alpha_{\iota,3}^{(q)}\left(\tilde{F}_G^{(q)}/\tilde{R}_G^{(q)}\right) + \alpha_{\iota,4}^{(q)}\right)\right.\\ &\left.-\Psi(\wp,\hat{\zeta}^{(q)})\right)^2, \forall \iota \in \{1,2,3\}\end{aligned} \tag{18}$$

In Eq. (18), the inner "$\max$" operator automatically selects the most suitable hyperplane closet to each evaluation point, and the outer "$\min$" operator guarantees the global accuracy between pre-selected hyperplane and all evaluation points. In the term of problem solving, (18) can be solved by directly calling `Ipopt` solver or reformulating it as its recursive form and subsequently invoking existing MIP solver, e.g., `Gurobi`, `IBM CPLEX`, etc. More detailed on the calculation of (18) can be found in [14, 15]

Substituting $\hat{\Psi}(\wp,\hat{\zeta}^{(q)})$ derived from Eq. (18) into Eq. (16), a convex form of (16) can be obtained as follows,

$$\begin{aligned}&\alpha_{\iota,1}^{(q)}H_{\text{equ}} + \alpha_{\iota,2}^{(q)}D_{\text{equ}} + \alpha_{\iota,3}^{(q)}\left(F_G/R_G\right) + \alpha_{\iota,4}^{(q)}\\ &\leq \Delta f_{\max} + M z_q; \sum\nolimits_{q=1}^{NQ}\pi_q z_q \leq \epsilon, \forall q, \iota \in \{1,2,3,4\}\end{aligned} \tag{19a}$$

IV. Proposed UC model:CCFDUC

This section presents a comprehensive articulation of CCFDUC in this work. The form of the suggested UC model can be constructed by adding frequency dynamic restrictions from section III into the baseline UC problem.

$$\begin{aligned}\min\sum\nolimits_{t=1}^{NT}&\left[\sum\nolimits_{g=1}^{NG}\left(c_g^{\text{up}}u_{g,t} + c_g^{\text{dw}}u_{g,t} + c_g^{\text{nl}}u_{g,t}\right) + \right.\\ &\left.\sum\nolimits_{s=1}^{NS}\pi_s\left(\sum\nolimits_{g=1}^{NG}\sum\nolimits_{z=1}^{NZ}F_g^z\left(p_{g,s,t}^{(z)}\right) + e^{\text{VoLL}}\sum\nolimits_{d=1}^{ND}p_{d,s,t}\right)\right]\end{aligned} \tag{20a}$$

$$s.t., \sum\nolimits_{m=\lceil t-T_g^{\text{su}}+1,0\rceil}^{t} u_{g,m} \leq x_{g,t}, \forall g,t \tag{20b}$$

$$\sum\nolimits_{m=\lceil t-T_g^{\text{sd}}+1,0\rceil}^{t} u_{g,m} \leq 1 - x_{g,t}, \forall g,t \tag{20c}$$

$$x_{g,t} - x_{g,t-1} = u_{g,t} - v_{g,t}, \forall g,t \tag{20d}$$

$$u_{g,t} + v_{g,t} \leq 1, \forall g,t \tag{20e}$$

$$\begin{aligned}&\sum\nolimits_{g=1}^{NG}p_{g,s,t} + \sum\nolimits_{w}^{NW}\left(p_{w,s,t} - \Delta p_{w,s,t}\right) + \sum\nolimits_{e=1}^{NE}\left(p_{e,s,t}^{\text{dis}} - \right.\\ &\left.\Delta p_{e,s,t}^{\text{ch}}\right) - \sum\nolimits_{d=1}^{ND}\left(p_{d,t} - \Delta p_{d,s,t}\right) = 0, \forall g,w,e,s,t\end{aligned} \tag{20f}$$

$$\begin{aligned}&|\sum\nolimits_{b=1}^{NB}\kappa_{l,b}\left(\sum\nolimits_{g=1}^{NG_b}p_{g,s,t} + \sum\nolimits_{w=1}^{NW_b}\left(p_{w,s,t} - \Delta p_{w,s,t}\right) + \right.\\ &\sum\nolimits_{e=1}^{NE_b}\left(p_{e,s,t}^{\text{dis}} - \Delta p_{e,s,t}^{\text{ch}}\right) - \\ &\left.\sum\nolimits_{d=1}^{ND}\left(p_{d,t} - \Delta p_{d,s,t}\right)\right)| \leq p_i^{\max}, \forall t,l,s\end{aligned} \tag{20g}$$

$$\begin{aligned}p_{g,s,t} - p_{g,s,t-1} &\leq \gamma_g^{\text{ru}}x_{g,t} + p_g^{\max}(1 - x_{g,t}) + \\ &\gamma_g^{\text{su}}(x_{g,t} - x_{g,t-1}), \forall g,s,t\end{aligned} \tag{20h}$$

$$\begin{aligned}p_{g,s,t-1} - p_{g,s,t} &\leq \gamma_g^{\text{rd}}x_{g,t} + p_g^{\max}(1 - x_{g,t-1}) + \\ &\gamma_g^{\text{sd}}(x_{g,t-1} - x_{g,t}), \forall g,s,t\end{aligned} \tag{20i}$$

$$\begin{aligned}&\sum\nolimits_{g=1}^{NG}\left(x_{g,t}p_g^{\max} - p_{g,s,t}\right) + \sum\nolimits_{e=1}^{NE}\left(p_{e,s,t}^{\text{dis, max}} - p_{e,s,t}^{\text{dis}}\right) \geq \\ &\varrho_1^+\sum\nolimits_{d=1}^{ND}p_{d,t} + \varrho_2^+\sum\nolimits_{e=1}^{NE}p_{w,s,t}^{\max}, \forall s,t\end{aligned} \tag{20j}$$

$$\sum\nolimits_{g=1}^{NG}\left(p_{g,s,t} - x_{g,t}p_g^{\min}\right) + \sum\nolimits_{e=1}^{NE}\left(p_{e,s,t}^{\text{ch, max}} - p_{e,s,t}^{\text{ch}}\right) \geq$$

$$\varrho_1^-\sum_{d=1}^{ND}p_{d,t}+\varrho_2^-\sum\nolimits_{e=1}^{NE}p_{w,s,t}^{\max},\forall s,t \tag{20k}$$

$$x_{g,t}p_g^{\max}\le p_{g,s,t}\le x_{g,t}p_g^{\max},\forall s,t \tag{20l}$$

$$0\le\Delta p_{w,s,t}\le p_{w,s,t}^{\max},\forall w,s,t \tag{20m}$$

$$0\le\Delta p_{d,s,t}\le p_{d,t},\forall d,s,t \tag{20n}$$

$$0\le p_{e,s,t}^{\text{dis}}\le u_{e,s,t}^{\text{dis}}p_e^{\text{dis,max}},\forall e,s,t \tag{20o}$$

$$0\le p_{e,s,t}^{\text{ch}}\le u_{e,s,t}^{\text{ch}}p_e^{\text{ch,max}},\forall e,s,t \tag{20p}$$

$$q_{e,s,t+1}=q_{e,s,t}+\left(\eta_e^{\text{ch}}p_{e,s,t}^{\text{ch}}-\eta_e^{\text{dis}}p_{e,s,t}^{\text{dis}}\right)\Delta t,\forall e,s,t \tag{20q}$$

$$q_{e,s,t=1}=q_{e,s,t=NT},\forall e,s,t \tag{20r}$$

$$u_{e,s,t}^{\text{dis}}+u_{e,s,t}^{\text{ch}}\le 1,\forall e,s,t \tag{20s}$$

$$p_{g,s,t}=\sum\nolimits_{z=1}^{NZ}p_{g,s,t}^{(z)},\forall z,g,s,t \tag{20t}$$

$$0\le p_{g,s,t}^{(z)}\le x_{g,t}p_{g,s,t}^{\max},\forall z,g,s,t \tag{20u}$$

$$F_g^z\left(p_{g,s,t}^{(z)}\right)\ge k_g^{(z)}p_{g,s,t}^{(z)}+x_{g,t}\omega_g^{(z)},\forall z,g,s,t \tag{20v}$$

$$\sum\nolimits_{g=1}^{NG}M_gx_{g,t}\ge\Delta p_{\text{step}}/RoCoF_{\max}-\sum\nolimits_{w=1}^{MW}M_w,\forall t \tag{21a}$$

$$\sum\nolimits_{g=1}^{NG}\left(p_g^{\max}-p_{g,s,t}\right)\ge K_GR_G^{-1}\Delta f_{\max},\forall s,t \tag{21b}$$

$$\sum\nolimits_{w=0}^{NG}(p_w^{\max}-p_{w,s,t})\ge\Delta f_{\max}/R_W\ ,\forall s,t \tag{21c}$$

$$\sum\nolimits_{e=0}^{NE}\left(p_e^{\max}-p_{e,s,t}^{\text{dis}}\right)\ge\Delta f_{\max}/R_E\ ,\forall s,t \tag{21d}$$

$$\sum\nolimits_{q=1}^{NQ}\pi_q\phi_{q,t}\le\epsilon,\forall q,t \tag{21e}$$

$$\alpha_{\iota,1}^{(q)}H_{\text{equ},t}+\alpha_{\iota,2}^{(q)}D_{\text{equ},t}+\alpha_{\iota,3}^{(q)}\left(F_{G,t}/R_{G,t}\right)+\alpha_{\iota,4}^{(q)} \le\Delta f_{\max}+M\phi_{q,t},\forall q,\iota\in\{1,2,3,4\} \tag{21f}$$

$$M_{\text{equ},t}=p_{\text{base}}^{-1}\left(\sum\nolimits_{g=1}^{NG}\left(M_gp_g^{\max}\right)x_{g,t}+\sum\nolimits_{w=1}^{NW}(M_wp_{w,s,t})\right) \tag{21g}$$

$$D_{\text{equ},t}=p_{\text{base}}^{-1}\left(\sum\nolimits_{g=1}^{NG}\left(D_gp_g^{\max}\right)x_{g,t}+\sum\nolimits_{w=1}^{NW}\left(R_w^{-1}p_{w,s,t}\right)+\sum\nolimits_{e=1}^{NE}\left(R_e^{-1}p_e^{\max}\right)\right),\forall t \tag{21h}$$

$$F_{G,t}=p_{\text{base}}^{-1}\sum\nolimits_{g=1}^{NG}\left(K_gF_gR_g^{-1}p_g^{\max}\right)x_{g,t},\forall t \tag{21i}$$

$$R_{G,t}^{-1}=p_{\text{base}}^{-1}\sum\nolimits_{g=1}^{NG}\left(K_gR_g^{-1}p_g^{\max}\right)x_{g,t},\forall t \tag{21j}$$

$$R_{W,t}^{-1}=p_{\text{base}}^{-1}\sum\nolimits_{w=1}^{NW}\left(K_wR_w^{-1}p_{w,s,t}\right),\forall t \tag{21k}$$

$$R_{E,t}^{-1}=p_{\text{base}}^{-1}\sum\nolimits_{e=1}^{NE}\left(K_eR_e^{-1}p_e^{\max}\right),\forall t \tag{21l}$$

$$x_{g,t}\in\{0,1\},u_{g,t}\in\{0,1\},v_{g,t}\in\{0,1\},u_{e,s,t}^{\text{dis / ch}}\in\{0,1\} \tag{21m}$$

$$p_{g,s,t}\in\mathbb{R}^+,\Delta p_{w,s,t}\in\mathbb{R}^+,\Delta p_{d,s,t}\in\mathbb{R}^+,p_{g,s,t}^{\text{dis / ch}}\in\mathbb{R}^+ \tag{21n}$$

In the above formulation, Eqs. (20a) -(20v) define a baseline UC model, [15], Eqs. (21a)-(21l) denote frequency dynamic constraints in line with the actual frequency requirements. It worth noting that the newly redesigned nadir constraint (21e)-(21f) is based on BF-SFR which is significantly different with previous SFR approach or UC studies.

Next, Eq. (20a) denotes the objective of the UC problem, it seeks to minimize the overall cost associated with conventional generators. Constraints (20b) and (20c) specify the minimum up-time and down-time requirements for all conventional generators, respectively. Constraints (20d) and (20e) ensure the coherence of the shut-up/shut-down choice and the operational condition of any individual conventional generator. Constraint (20f) delineates the constraint pertaining to the online power balance restriction. Constraint (20g) denotes the constraint that represents the capacity of each transmission line. Constraints (20h) and (20i) ensure the compliance of the up-forward and down-forward ramping rates of each conventional generator throughout consecutive dispatching periods. Constraints (20j) and (20k) delineate the stipulation for the provision of up-forward and down-forward spinning reserve capacity generated by individual conventional generator. Constraint (20l) indicate the minimize/minimize power of multiple generators. Constraints (20m) and (20n) introduce limitations on the amount of abandoned wind output and forced-load-curtailment. Constraints (20o) and (20p) specify the binding of charging/discharging capacities for each BESS. Constraint (20q) signifies a coupling restriction between the energy of BESS and the corresponding discharging/charging decisions. Constraint (20r) establishes a periodic energy regulation policy for each BESS. Constraint (20s) underscores the exclusiveness of the discharging and charging choices made by each BESS. Constraint (20t) enforces that the output of individual conventional generator is equal to the total of the segmented outputs generated on each block of its linearized fuel cost curve. Constraint (20u) enforces the binding restriction of each segmented output. Constraint (20v) specifies the operational cost associated with each particular conventional generator. Constraints (21a) to (21f) correspond to the RoCoF, nadir and FCR constraint, which are used to mitigate the impact against a specific power disturbance. It is noteworthy to mention that the original chance nadir constraint (15) has been transformed into a collection of convex constraints using SAA. Constraints (21g) to (21j) declare aggregated values of the system inertia constant, damping constant, fraction of total output generated by conventional reheated generators and droop coefficient, respectively. Constraints (21k) and (21l) represent the aggregated droop coefficient of wind farms and BESS. Constraint (21m) and (21n) describe the feasible region of binary decision variables and continuous decision variables, respectively.

V. Case Studies and Analysis

This section uses two cases, including a modified version of IEEE 6-bus test system and IEEE 118-bus test system to check the validness and scalability of CCFDUC in this work. Both experiments are carried out in Julia (version 1.9.2) / JuMP (version 1.13.0) environment, [26], using Gurobi (version 10.0) on a Dell Precision T5820 Tower workstation with Intel(R) Xeon(R)W-2133 CPU@3.60 GHz & RAM 12.0GB.

A. Modified IEEE 6 bus test system

The system comprises 6 buses, 7 lines, and 3 conventional generators. Besides, the system contains 4 wind farms located on bus 1 and 2 BESS at bus 2. The installed capacity generated by conventional generators is $240MW$; the capacity of each wind farm is $50MW$, and the similar one of each BESS is $50MW$. In such a condition, the percentage of wind farms for the peak load is 29.40%. Besides, a standardized value method is used to construct the boundary condition: the baseline power is $100MW$ and the baseline frequency is $50Hz$. Other parameters including units' shut-up / shut-down time, ramping rate restrictions, maximum/minimum outputs, demand consumption and network topology parameters can be referred to [6]. The chronological load curve and the stochastic scenarios of wind farms are illustrated in Fig. 3. Table. II sorts out the frequency regulation parameters of different generators.

Table II
FREQUENCY DYNAMIC PARAMETERS OF DIFFERENT GENERATORS.

Gen.NO.	Type	H	K	F	R	D	T
1	Thermal	7	0.9	0.04	0.04	0.06	8
2	Thermal	5.5	0.95	0.03	0.03	0.1	8
3	Thermal	3.5	0.98	0.25	0.05	0.18	8
4	Wind farms	—	1	—	0.04	—	0.06
5	Wind farms	6	1	—	—	0.6	0.06
6	BESS	—	0.5	—	0.04	—	0.06

$A.1.\ Simulator\ v.s.\ BF-SFR(the\ proposed)$

A simulator on SFR at a MATLAB 2022a simulink environment is used to compare the feasibility of BF-SFR. Both grid-connected generators with similar types are abstracted as an aggregated generators participating in frequency regulations. Especially, a step type mismatch power Δp_{step} (approximately equal to 20% loads) is simulated using simulator and compared against BF-SFR. Two residual settings are selected to compare their impacts on the frequency dynamic: i): a little residual condition: it assumes that the obtained power derivation should be derivative from real signals with a minor probability, and the residual is assumed to be followed a normal distribution i.e., $\langle \varpi, \sigma \rangle \in N(0, 0.001)$ where $\langle \varpi, \sigma \rangle$ denotes the expectation and standard deviation of measured UV errors. In such a setting, the impact of UV from wind farm power fluctuation could be neglect; ii): a relatively large residual condition: this case assumes that the power derivations follow a normal distribution as $\langle \varpi, \sigma \rangle \in N(0, 0.01)$ due to the backward of measuring terminal devices or a significant power changing of wind farms. Fig. 4 shows different SFR curves solved by BF-SFR/simulator under two residual conditions mentioned before. For a comprehensive comparison of BF-SFR / simulator with different grid-connected generator types and different UV settings, the impact of introducing wind farms into frequency supporting is considered in Fig. 4. Accordingly, "Case 1" in Fig. 4 represents the frequency dynamic curve without ("w/o") considering frequency supporting of wind farms, whilst "Case 2" represents the curve with ("w/") considering the frequency regulation produced by wind farms.

In Fig. 4, both curves of $\Delta f(t)$ solved by simulator and BF-SFR are close, it demonstrates the validness of the BF-SFR. Specially, the maximum value of $\Delta f(t)$ in Case 2 no matter in Fig. 4(a) or Fig. 4(b) is much minor than its counterpart in Case 1, leading to smoother frequency dynamic response and better frequency security performance. It declares that the security of frequency dynamic would be improved if additional frequency supporting generated by wind farms is integrated. Furthermore, the changing of $\Delta f(t)$ in a large residual setting is significantly intense than that in the minor residual situation. It depicts that the UV characteristic of intermittent resource cannot be ignored, especially in the low-inertia grid caused by a high share of renewable resource.

Another feature of BF-SFR different from previous analytical SFR, time-discretized SFR and simulator is that BF-SFR can not only capture the chronological changing trajectory of $\Delta f(t)$ with multiple control schemes but also provide its posterior information, e.g., CDF, PDF, $1/2$ and $1/4$ quartiles, etc. Fig. 5 shows the posterior probabilistic density distribution and associated quartile level of $\Delta f(t)$ under different residual conditions. Fig. 6 illustrates the union probabilistic density distribution of $\Delta f(t)$ in the entire simulation analysis interval $(0s - 30s)$.

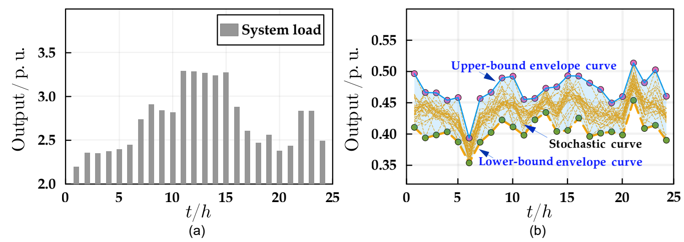

Figure 3. System load curve in (a) and stochastic wind power scenarios in (b) in the modified IEEE 6 bus test system.

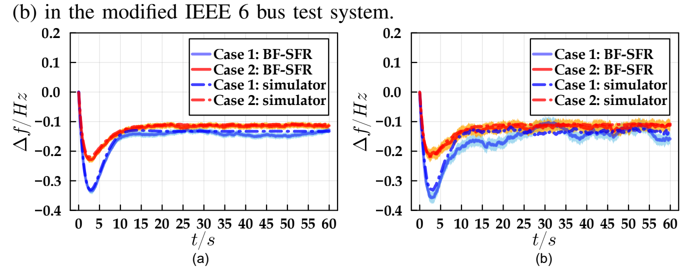

Figure 4. Frequency curves of BF-SFR/simulator with little residual setting in (a) and large residual condition in (b).

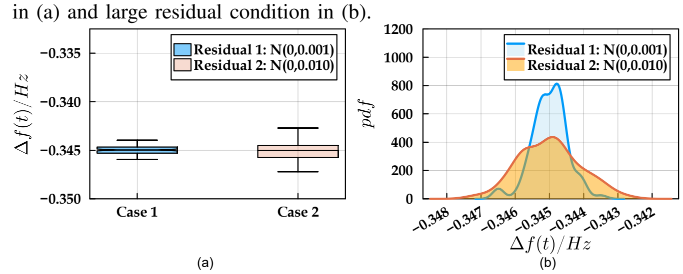

Figure 5. Statistical $1/2$ and $1/4$ quantile level in (a) and posterior probabilistic density distribution in (b) of $\Delta f(t)$ at time-to-nadir under different residual conditions.

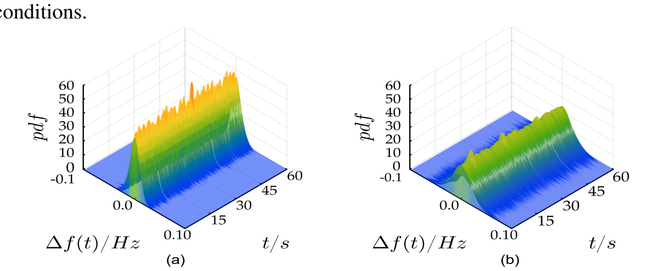

Figure 6. Union probabilistic density distribution of $\Delta f(t) : t \in [0s, 60s]$ under minor residual condition $N(0, 0.001)$ in (a) and relatively large residual condition $N(0, 0.01)$ in (b).

In Fig. 6, the probabilistic density distribution of $\Delta f(t)$ converges to a certain distribution at the end of the simulation time, it reflects that SFR in this case has a stationary probabilistic distribution. Especially, the distribution of $\Delta f(t)$ at the minor residual condition is sharper than its counterpart at the relatively large residual condition, leading to reduced uncertain risks. Hence, the security of frequency dynamic can be well maintained if limited UV factors are incorporated. On the contrary, if there are a great number of UV derivation from converters introducing into the grid, the possible value of $\Delta f(t)$ would become broader, it leads to a difficult prediction and re-dispatching issue. Therefore, the binding of SFR at a significant UV residual condition is essential by defining the $1/2$ quartile and $1/4$ quartile filtered from sampled frequency dynamic $filters$ (see Fig. 5(b)) through the proposed BF-SFR method.

$A.2.\ TUC\ v.s.\ FDUC\ v.s.\ CCFCUC(the\ proposed)$

To check the performance of introducing frequency dynamic requirements via BF-SFR into UC problems, TUC, [27], and recent previous frequency-dynamic UC (FDUC) model, [15], are applied as benchmarks and sequentially compared with the

Table III
SCHEDULED COST INCLUDING UNITS' SHUT-ON COST, SHUT-DOWN COST AND FUEL COST PRODUCED BY TUC/FDUC/CCFDUC.

Models	units' shut-up cost	units' shut-down cost	units' fuel cost
TUC	1.17×10^4\$	6.10×10^4\$	4.082×10^5\$
FDUC	9.10×10^3\$	0	4.810×10^5\$
CCFDUC	9.10×10^3\$	0	4.810×10^4\$

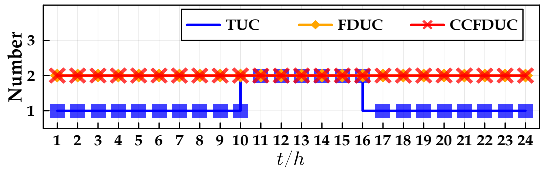

Figure 7. Cumulative numbers of activated conventional generators through TUC/FDUC/CCFDUC in the modified IEEE 6 bus test system.

proposed CCFDUC model (see (20a) – (21n)). The differences between FDUC and CCFDUC lie in the construction of frequency nadir constraints. In the context of boundary conditions representing frequency dynamic requirements, the limit of RoCoF and nadir are, respectively, $1.0Hz/s$ and $0.5Hz$. The residual condition describing UV derivation during frequency dynamic processes is $\langle \varpi, \sigma \rangle \in N(0, 0.001)$. Accordingly, Fig. 10 compares the calculated scheduled costs and commitment results produced by TUC, FDUC and CCFDUC.

In Table III, both schedule costs solved by FDUC / CCFDUC are close, it demonstrates the feasibility of integrating SFR via BF-SFR into UC. In comparison with the latter two frequency-dynamic UC models, all units' fuel cost through TUC is minor than that through FCUC and CCFDUC, while units' shut-on/shut-down costs using TUC are higher than its counterpart. It reflects that more conventional generators run at a bit uneconomical level in FDUC/CCFDUC over the entire dispatching horizons but only similar activated generators exist in TUC during limited peak-load periods for the sake of active power balance restriction. Note that FDUC/CCFDUC consider the frequency supporting benefit from wind farms and BESS, these grid-connected devices must preserve a certain frequency containment reserve for keeping adequate frequency regulation capacities. In this perspective, these grid-connected generators other than conventional generators through FDUC/CCFDUC run at a lack of economic point compared with the one via TUC. Therefore, activated conventional generators through FDUC/CCFDUC must produce more output for keeping active power balance. The comparison of accumulated output of conventional generators through TUC ($24.976MW$), FDUC ($26.404MW$ in Fig. 8(a)) and CCFDUC ($26.402MW$ in Fig. 8(b)) demonstrated this explanation.

Furthermore, the number of activated conventional generators in Fig. 7 through TUC is broadly minor than that through FDUC/CCFDUC expect for limited peak-load periods. This issue can be more pronounced at 1h−9h and 17h−24h. During these period internals, only G1 is activated through TUC but G1 and G2 keep online through FDUC/CCFDUC. It is because that the scheduled generators produced by frequency-dynamic UC schemes must dispatch more conventional generators to keep online in order to maintain an appropriate inertia level, but this issue is implicitly ignored in the TUC. Therefore, a conclusion can be drawn that the impact of incorporating SFR problems into UC would bring more generators to keep online to maintain a suitable inertia degree and frequency supporting

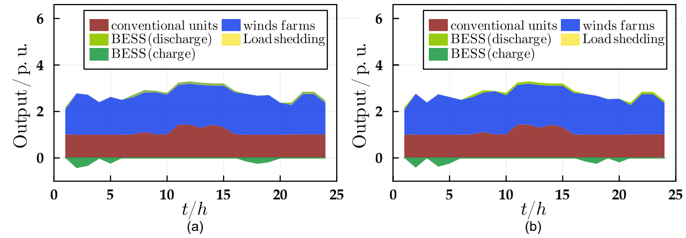

Figure 8. Active power balance curves of calculated scheduled commitments produced by TUC/FDUC/CCFDUC in the modified IEEE 6 bus test system.

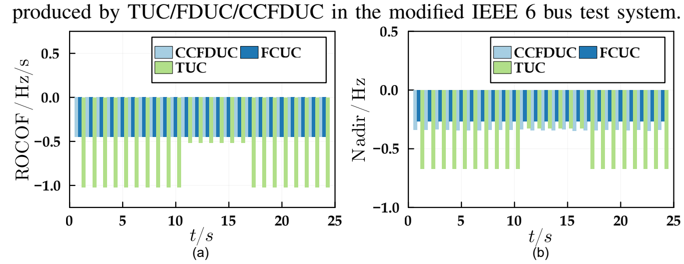

Figure 9. The distribution of RoCoF in (a) and nadir in (b) corresponding to scheduled commitments through TUC/FDUC/CCFDUC in the modified IEEE 6 bus test system.

capacity, resulting in safer frequency dynamic response against mismatch power disturbance.

To compare the effectiveness of introducing SFR via BF-SFR into UC studies, Fig. 9 statistics chronological curves of RoCoF and frequency nadir regarding scheduled commitments produced by TUC/FDUC/CCFDUC. It can be found that the scheduled results through FDUC/CCFDUC satisfy predefined frequency dynamic thresholds, but TUC does not. As a result, the necessity of introducing SFR into UC issues can be well demonstrated.

Nevertheless, the values of RoCoF/nadir with similar commitment results in Fig. 9 solved by TUC and FDUC/CCFDUC still has a slight difference. This phenomenon can be found and highlighted at 11h−16h in Fig. 9. At these periods, both calculated commitment results through TUC/FDUC/CCFDUC are similar, i.e., G1/G2 is on and G3 is off. In spite of this, there still exist several improvements of RoCoF and frequency nadir in the scheduled results produced by FDUC/CCFDUC. It is because that wind farms and BESS are took into account to participate into frequency supporting within FDUC/CCFDUC but TUC does not. Since the maximum of RoCoF is inversely proportional to the system inertia according to eq.(10), thereby, the integrating of wind farm with VI control can increase the system inertia and reduce RoCoF, and the benefit of droop-controlled wind farm and BESS results in an increase in available frequency regulation capacity, facilitating smoother frequency response and adequate frequency reserve inspired by Eqs. (11), (14) and (19). The specific frequency curve of scheduled commitments via TUC/FDUC/CCFDUC at $t = 10$h in Fig. 10 verified this explanation. It can be found in Fig. 10 that the frequency nadir using FDUC/CCFDUC is smaller than that through TUC, leading to smoother frequency response curve and adequate frequency reserve.

Next, Fig. 14 explores the evolution of scheduling results via CCFDUC with various residual settings. The scheduling cost will rise with the increase of external residual influences; in the meantime, the risk of abandoned wind output and forced-load-curtailment for preventing power-balance violation would also increase. As the residual changes within a relatively low internal between 1×10^{-3} and 2×10^{-3}, the scheduling cost

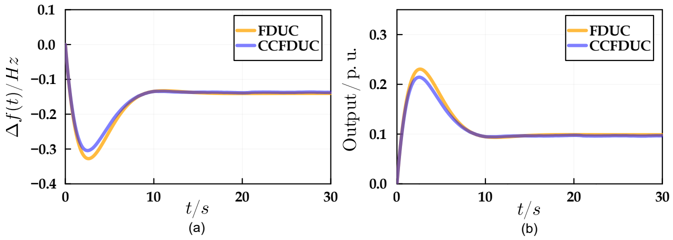

Figure 10. Frequency dynamic curves in (a) and additional output in (b) through TUC/FDUC/CCFDUC in the modified IEEE 6 bus test system.

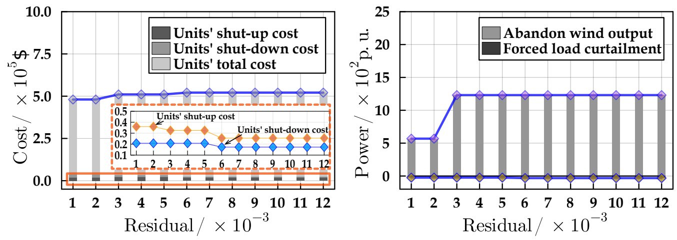

Figure 11. Change of scheduled cost and abandoned power through CCFDUC with different residual settings.

Table IV
SCHEDULED COST INCLUDING UNITS' SHUT-ON COST, SHUT-DOWN COST AND FUEL COST PRODUCED BY TUC/FDUC/CCFDUC.

Models	units' shut-up cost	units' shut-down cost	units' fuel cost
TUC	1.17×10^4\$	6.10×10^4\$	4.082×10^5\$
FDUC	9.10×10^3\$	0	4.810×10^5\$
CCFDUC	9.10×10^3\$	0	4.810×10^4\$

through CCFDUC increases slowly, it indicates that the minor residual has a negligible effect on the final scheduling result. However, when the residual changes from 2×10^{-3} to 3×10^{-3}, the scheduling results have a relatively obvious increase. This is mainly because some inactivated units in previous minor residual settings are required to be shut up in such condition. And, when the residual continuously increases from 3×10^{-3} to 10×10^{-3}, due to the fact that more units keep active and the system inertia has maintained at a relatively high degree, hence, the scheduling cost and power abandonment growth are maintained at a relatively flat curve. From this perspective, the necessity and significance of introducing UV into UC studies is well clarified.

B. Modified IEEE 118 test system

The system contains 54 conventional units, 186 lines and 91 loads. The cumulative installed capacity of conventional generators amounts to $7,220$MW, while the peak-load reaches $5,516.08$MW. In addition, the system encompasses 3 wind farms $\langle$G55, G56, G57$\rangle$, located on buses $\langle 24, 25, 26\rangle$, respectively. Each wind farm has a installed capacity of 500MW. The criteria of RoCoF and nadir are 2.0Hz/s and 0.5Hz. More boundaries about all units' physical operation parameters and frequency parameters can be found in Ref. [28]. an increase of load (approximately 10% of peak load) is chosen to define a representative power disturbance. Table IV presents the dispatching costs through TUC/FDUC/CCFDUC. Meanwhile, Fig. 14 statistics the number of activated generators over the entirety of the dispatching periods using various UC models. Figure 19 compares the distribution of RoCoF and nadir in relation to various obtained scheduling commitments through TUC/FDUC/CCFDUC.

In table IV, the scheduled costs and activated units' number through FDUC/CCFDUC exhibit a bit greater value compared to those through TUC. The major distinction for this difference

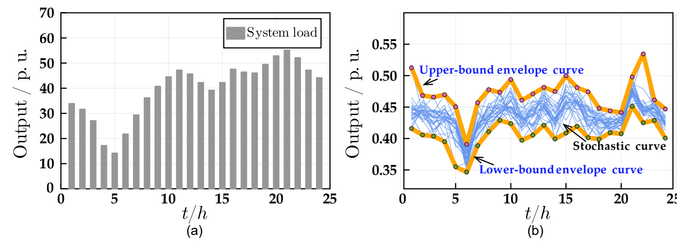

Figure 12. Load curve in (a) and stochastic wind power scenarios in (b) in the modified IEEE 118 bus test system.

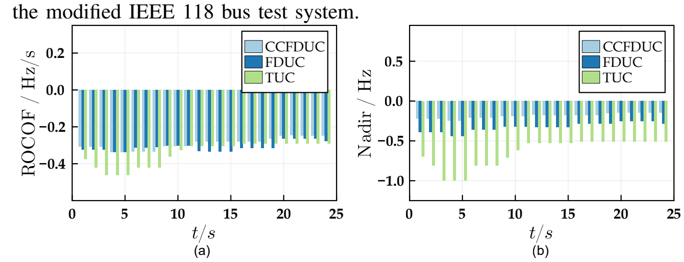

Figure 13. The distribution of RoCoF in (a) and nadir in (b) corresponding to scheduled commitments through TUC/FDUC/CCFDUC in the modified IEEE 118 bus test system.

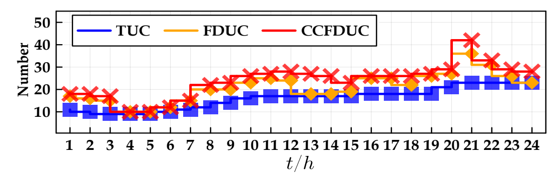

Figure 14. Cumulative numbers of activated conventional generators through TUC/FDUC/CCFDUC in the modified IEEE 118 bus test system.

lies in an reconsideration of frequency dynamic requirements. Such units with adequate frequency regulation capabilities are prioritized in FDUC/CCFDUC, and this selection of active generators utilizing this logicism is particularly evident during low-load periods (see scheduling periods $3h-6h$ in Fig. 14). It can be observed in Fig. 14 that the number of activated generators through TUC is lower compared to those activated through FDUC/CCFDUC. It is because that the system necessitates a restricted number of conventional generators for maintaining power balance. Nevertheless, these activated generators behind TUC may not provide an adequate level of the system inertia, resulting in a limited capability for these units to keep frequency dynamic security. Therefore, supplementary generators are compelled to remain operational with a reduced output so as to sustain the required amount of the system inertia.

The number of activated generators using FDUC and CCFDUC exhibits similarity, hence reinforcing the validity of the suggested model. Nevertheless, there are still minor disparities in the scheduled results employed by these two models under restricted dispatching intervals. For instance, at a specific time of 21h, the number of conventional generators determined by FDUC is 36, but the number determined using CCFDUC is 40. The discrepancy between FDUC/CCFDUC can be attributed to the altered formulation of frequency restrictions that pertain to frequency dynamic security. The frequency constraints in FDUC stem from a analytical SFR approach, [1, 2, 6], with an implicitly free UV assumption. In contrast, the frequency limits in CCFDUC are based on the BF-SFR that incorporates specific UV assumptions. Given the consideration of both the advantages of additional frequency control through converters and the associated fluctuations that may lead to UV residuals, CCFDUC tends to strike a balance, and this balance involves maximizing the incorporation of converters to enhance system frequency security while mitigating the adverse impact of UV issues. Noted that the nadir exhibited by CCFDUC is superior to that of FDUC. This indicates a higher level of robustness in terms of scheduling obligations.

VI. Conclusion

This paper presents a novel designed BF-SFR method for encoding chance constraint frequency dynamic requirements in CCFDUC. The suggested model allows for the flexible consideration and extension of sophisticated frequency control schemes and stochastic residuals representing UV issues. The reformulation and linearization of the chance-constraint nadir constraints derived from BF-SFR is discussed. Two cases are carried out to check the validness and scalability of BF-SFR and FCUC methodologies.

Three findings indicate that: $i)$ The trajectory of frequency derivation solved by BF-SFR is consistent with the simulated curve, and the probabilistic distribution of frequency derivation under different residual settings can be well visualized through BF-SFR. $ii)$ Introducing frequency dynamic requirement in UC may increase the scheduled cost and the number of grid-connected generators, nevertheless, the resulting scheduling result from FDUC/CCFDUC is beneficial in terms of ensuring system frequency security. $iii)$ CCFDUC has better performance in balancing the participation of converters in frequency support and avoiding excessive UV perturbations.

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